SATGURU STUDY CENTRE

Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

UNIT TEST-1, JULY - 2024

BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 1.30 Hours] **[Max. Marks : 50**

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark.

10x1=10

1. The value of $\begin{vmatrix} 2x+y & x & y \\ 2y+z & y & z \\ 2z+x & z & x \end{vmatrix}$ is
   (a) xyz (b) $x+y+z$ (c) $2x+2y+2z$ (d) 0
2. If $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix}$ then $\begin{vmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{vmatrix}$ is
   (a) Δ (b) $-\Delta$ (c) 3Δ (d) -3Δ
3. If A is square matrix of order 3 then $|kA|$ is
   (a) $k|A|$ (b) $-k|A|$ (c) $k^3|A|$ (d) $-k^3|A|$
4. adj (AB) is equal to
   (a) adjA adjB (b) adj^T adjB B^T (c) adjB adjA (d) adj^T adj^T
5. The number of Hawkins-Simon conditions for the viability of an input-output analysis is
   (a) 1 (b) 3 (c) 4 (d) 2
6. If A and B non-singular matrix then, which of the following is incorrect?
   (a) $A^2 = I$ implies $A^{-1} = A$ (b) $I^{-1} = I$
   (c) If $AX = B$ then $X = B^{-1}A$ (d) If A is square matrix of order 3 then $|adj\,A| = |A|^2$
7. The value of n, when $np_2 = 20$ is
   (a) 3 (b) 6 (c) 5 (d) 4
8. The number of diagonals in a polygon of n sides is equal to
   (a) nC_2 (b) $nC_2 - 2$ (c) $nC_2 - n$ (d) $nC_2 - 1$
9. The greatest positive integer which divide $n(n+1)(n+2)(n+3)$ for all $n \in N$ is
   (a) 2 (b) 6 (c) 20 (d) 24
10. The number of permutation of n different things taken r at a time, when the repetition is allowed is
   (a) r^n (b) n^r (c) $\frac{n!}{(n-r)!}$ (d) $\frac{n!}{(n+r)!}$

PART – II

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question number 16 is compulsory

4x2=8

11. Find the inverse of the matrix $\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$
12. Solve by using matrix inversion method: $2x + 5y = 1$, $3x + 2y = 7$
13. Evaluate: (i) $8P_3$ (ii) $5P_4$
14. If $15C_{3r} = 15C_{r+3}$, find r
15. Expand using binomial theorem $\left(x + \frac{1}{x^2}\right)^6$
16. The technology matrix of an economic system of two industries is $\begin{bmatrix} 0.50 & 0.30 \\ 0.41 & 0.33 \end{bmatrix}$ Test whether the system is viable as per Hawkins Simon conditions.

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PART – III

4x3=12

1. **Answer any 4 questions**
2. **Each question carries 3 marks**
3. **Question number 22 is compulsory**

17. Evaluate: $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$

18. In an economy there are two industries P_1 and P_2 and the following table gives the supply and the demand position in crores of rupees.

Production sector	Consumption sector		Final demand	Gross output
P1	10	25	15	50
P2	20	30	10	60

Determine the outputs when the final demand changes to 35 for P_1 and 42 for P_2.

19. Resolve into partial fractions : $\dfrac{x-2}{(x+2)(x-1)^2}$

20. Using binomial theorem, evaluate $(101)^5$

21. Find the rank of the word 'CHAT' in dictionary.

22. If $(n+2)C_n = 45$, find n

PART – IV

4x5=20

1. **Answer all the questions**
2. **Each question carries 5 marks**

23. a) Prove that $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$

(OR)

b) Evaluate $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

24. a) Solve by using matrix inversion method: $3x - 2y + 3z = 8$; $2x + y - z = 1$, $4x - 3y + 2z = 4$.

(OR)

b) The sum of three numbers is 20 . If we multiply the first by 2 and add the second number and subtract the third we get 23 . If we multiply the first by 3 and add second and third to it,we get 46. By using matrix inversion method find the numbers.

25. a) Resolve into partial fraction: $\dfrac{x+4}{(x^2-4)(x+1)}$

(OR)

b) $1.2 + 2.3 + 3.4 + \dots + n(n+1) = \dfrac{n(n+1)(n+2)}{3}$, for all $n \in N$.

26. a) Show that the matrices $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} \frac{4}{5} & \frac{-2}{5} & \frac{-1}{5} \\ \frac{-1}{5} & \frac{3}{5} & \frac{-1}{5} \\ \frac{-1}{5} & \frac{-2}{5} & \frac{4}{5} \end{bmatrix}$ are inverse of each other.

(OR)

b) A Committee of 5 is to be formed out of 6 gents and 4 ladies. In how many ways this can be done when (i) atleast two ladies are included (ii) atmost two ladies are included